Part 1: Data Understanding

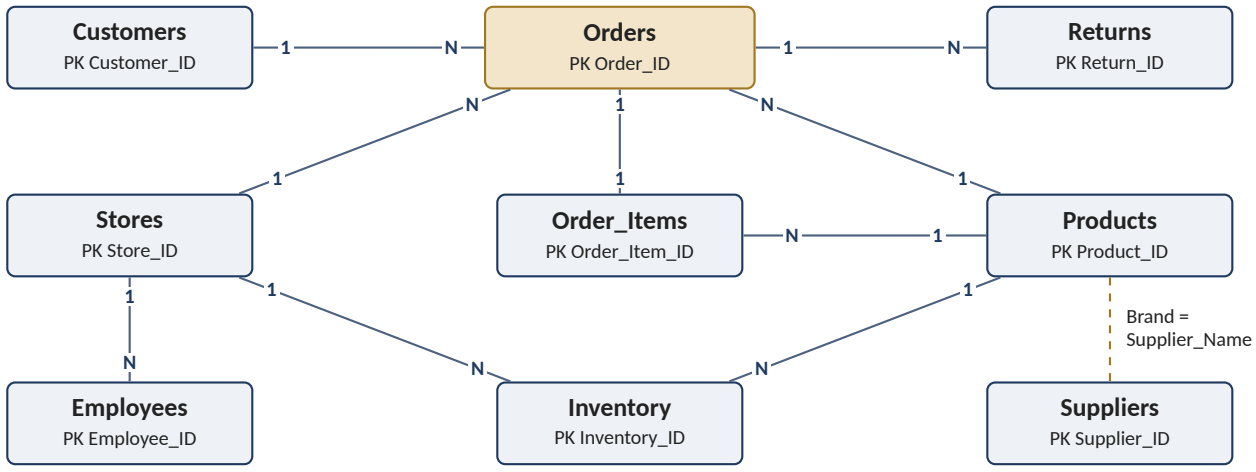
ShopSmart Retail Ltd • Retail Sales Analysis with Microsoft Excel

ShopSmart gave us **10 files**: nine data tables (all CSV) and one Data Dictionary workbook that explains what the columns mean. Before opening Excel properly, I went through every file to see what it holds and how the pieces connect.

What each dataset represents, and how it is keyed

Dataset	What it holds	Primary Key	Foreign Key(s)
Customers	Who buys: demographics, contact details and membership level (Bronze, Gold and so on)	`Customer_ID`	None
Products	The catalogue: product name, category, brand, cost price and selling price	`Product_ID`	None (links to Suppliers by `Brand`)
Stores	The physical branches: store name, city, state and region	`Store_ID`	None
Employees	Staff records: role, monthly salary and the store they work in	`Employee_ID`	`Store_ID`
Suppliers	Contact and location details of the brands that supply ShopSmart	`Supplier_ID`	None
Inventory	Current stock level, reorder level and restock date for each product in each store	`Inventory_ID`	`Product_ID`, `Store_ID`
Orders	The main transaction log: who bought, at which store, when, and the order status	`Order_ID`	`Customer_ID`, `Store_ID`, `Product_ID`
Order_Items	Line-level money details: quantity, unit price, discount and line total	`Order_Item_ID`	`Order_ID`, `Product_ID`
Returns	Refunded orders: the return reason and the amount refunded	`Return_ID`	`Order_ID`
Data Dictionary	Reference sheet that defines every column in the nine tables	Not a data table	Not applicable

How the tables relate



Reading the diagram: 1 means the "one" side and N the "many" side. Orders is the centre of the model, with Customers, Stores and Products feeding into it, and Returns hanging off it. Inventory sits between Products and Stores, and Employees hang off Stores. The dashed line is a match on a name, not an ID. I drew this diagram in LaTeX (TikZ) so the connections stay exact and sharp.

Two things I noticed

- **No Supplier_ID in Products.** The only way to reach a supplier is to match `Brand` in Products to `Supplier_Name` in Suppliers. I will use that as the lookup key later.
- **Risk of double counting revenue.** Orders and Order_Items each have 50,000 rows (one item per order), and both carry Quantity, Unit_Price and Discount. If I merge them and sum both, revenue doubles. I will pick one table as the source for every revenue formula.

Part 8: Business Questions

ShopSmart Retail Ltd • Answers drawn from the cleaned workbook

1	**Which product generated the highest revenue?**	Home Item 400, with **₦21,638,851**.
2	**Which store performed best?**	ShopSmart Lagos, with **₦1,232,192,532** in revenue.
3	**Which category sells the most?**	Beauty, with **₦1,762,500,500** in revenue.
4	**Which month generated the highest revenue?**	December, at **₦660,629,527**.
5	**Which customers spend the most?**	Joseph Smith leads with **₦12,009,782**. Nicole Smith and Mark Smith come next.
6	**Which products are frequently returned?**	Home Item 263 has the most returns (**11**). Beauty Item 220 and Beauty Item 262 follow with 10 each.
7	**Which supplier supplies the highest number of products?**	Three suppliers tie at **2 products each**: Dyer LLC Supplies, Garcia PLC and Jones Group. Most products come from a different supplier, since there are 489 suppliers in total.
8	**Which stores have low inventory?**	ShopSmart Enugu is lowest with only **3,349 items** on hand. ShopSmart Kano (4,708) and ShopSmart Awka (7,853) come after it.
9	**Which region performs best?**	The South Region, with **₦2,424,534,663** in total revenue.
10	**Which product category should management invest in?**	Groceries. Beauty earns a little more, but Groceries still brought in about **₦1.57 billion** and has the lowest return rate of any category at **4.57%**. That means more of its sales stay as sales, so it is the safer place to put money.

Part 9: Business Recommendations

ShopSmart Retail Ltd • Executive summary

Three key findings

1. **The South Region leads by a wide margin.** It brought in over ₦2.42 billion, and a big part of that comes from the Lagos branch.
2. **Beauty earns the most, but Groceries is close.** Beauty made ₦1.76 billion. Groceries is not far behind and has a much lower return rate.
3. **Sales peak at the end of the year.** Revenue climbs towards year end and reaches its highest point in December at ₦660.6 million, which points to strong holiday spending.

Three business recommendations

1. **Grow the grocery range.** Groceries has the lowest return rate (4.57%) and strong sales, so it is worth adding more grocery products and promoting them to our top-tier members.
2. **Learn from Lagos and copy what works.** Review how the Lagos branch sells and lays out its store, then share those practices with the weaker branches.
3. **Bundle the high-return items.** Products that get returned often, like Home Item 263, can be packaged with items that sell well. This helps clear the stock without taking a full loss.

One risk identified

Stockouts in some branches

ShopSmart Enugu has only 3,349 items on hand, the lowest of all stores. If it runs out of stock, shelves go empty, sales are lost and customers may stop coming back.

One opportunity identified

The year-end rush

December is the peak month, so there is room to plan for it earlier. Next year, ShopSmart can raise its marketing spend and run targeted Gold Membership offers in November to push revenue higher.